CAPSTONE PROGRESS REPORT

Project Title: Lane Detection Using Deep Learning
Course: DATS 6501 – Data Science Capstone

Summary

This project aims to design and implement a deep learning-based semantic segmentation system for lane detection in road images and video streams.

Since the proposal submission, the project has progressed from conceptual design to functional implementation of a working training pipeline and initial inference results on video data.

The system now supports:

- Image preprocessing pipeline
- Fully Convolutional Network (FCN) training
- Binary lane mask prediction
- Overlay visualization on test videos

The project is currently in the model refinement and robustness improvement phase.

Problem Revalidation

Lane detection remains challenging under:

- Variable lighting
- Curved roads
- Shadows
- Faded lane markings

Traditional rule-based methods fail in generalization.

Work Completed So Far

A. Data Engineering

- Image mask pairing verified

- Applied augmentation:
    - Horizontal flips
    - Rotations
    - Brightness adjustments
- Split into training / validation / testing

B. Model Development

- Implemented Fully Convolutional Network (FCN)
- Batch training with validation tracking
- Saved trained model (.h5 format)

C. Inference Pipeline

- Video frame extraction
- Lane overlay projection
- Annotated output video generation

D. Initial Results

- Model converges successfully
- Visual lane detection works on straight roads
- Moderate performance on curved roads

Quantitative Results

- Final Training Loss – 0.08
- Model forward-pass latency (per frame) – ~10–16 ms
- Pixel Accuracy – ~95%
- IoU – Not implemented yet (planned)
- FPS during inference – ~30 FPS

Visual Results (Before vs After Processing)

Before Processing (Original Frame)

The input frame is directly extracted from the driving video.
At this stage, no segmentation or post-processing has been applied.
The lane boundaries are visible but not highlighted or enhanced.

After Processing (Lane Detection Output)

This frame shows the output of the Fully Convolutional Network (FCN).
The predicted binary lane mask is:

- Resized back to full resolution
- Placed into the green channel
- Blended with the original frame

Next Development Milestones

This Session (Current)

Working FCN model - Done
Video inference demo - Done

Next Session

- Implement IoU & F1 metrics
- Improve loss function (Dice Loss / Focal Loss)
- Add confidence heatmap visualization
- Lane confidence scoring system
- Risk alert module